GG–G–590
August 11, 1961

FEDERAL SPECIFICATION

GONIOMETER, ORTHOPEDIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of orthopedic goniometer (Conzett design) suitable for measuring the degree of flexion and extension of joints.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

GG–I–526—Instruments, Dental and Surgical; General Specifications.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–12902—Packaging of Surgical, Dental and Veterinary Instruments, Diagnostic Equipment, and Related Items.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Protractor, fixed and movable arms, knob, and bushing.* The protractor, the fixed and movable arms, and the knob and bushing shall be fabricated of class 2, composition G steel conforming to the requirements of Federal Specification GG–I–526, except that the Rockwell hardness shall be B75 minimum.

3.1.2 *Shims.* The shims shall be fabricated of brass.

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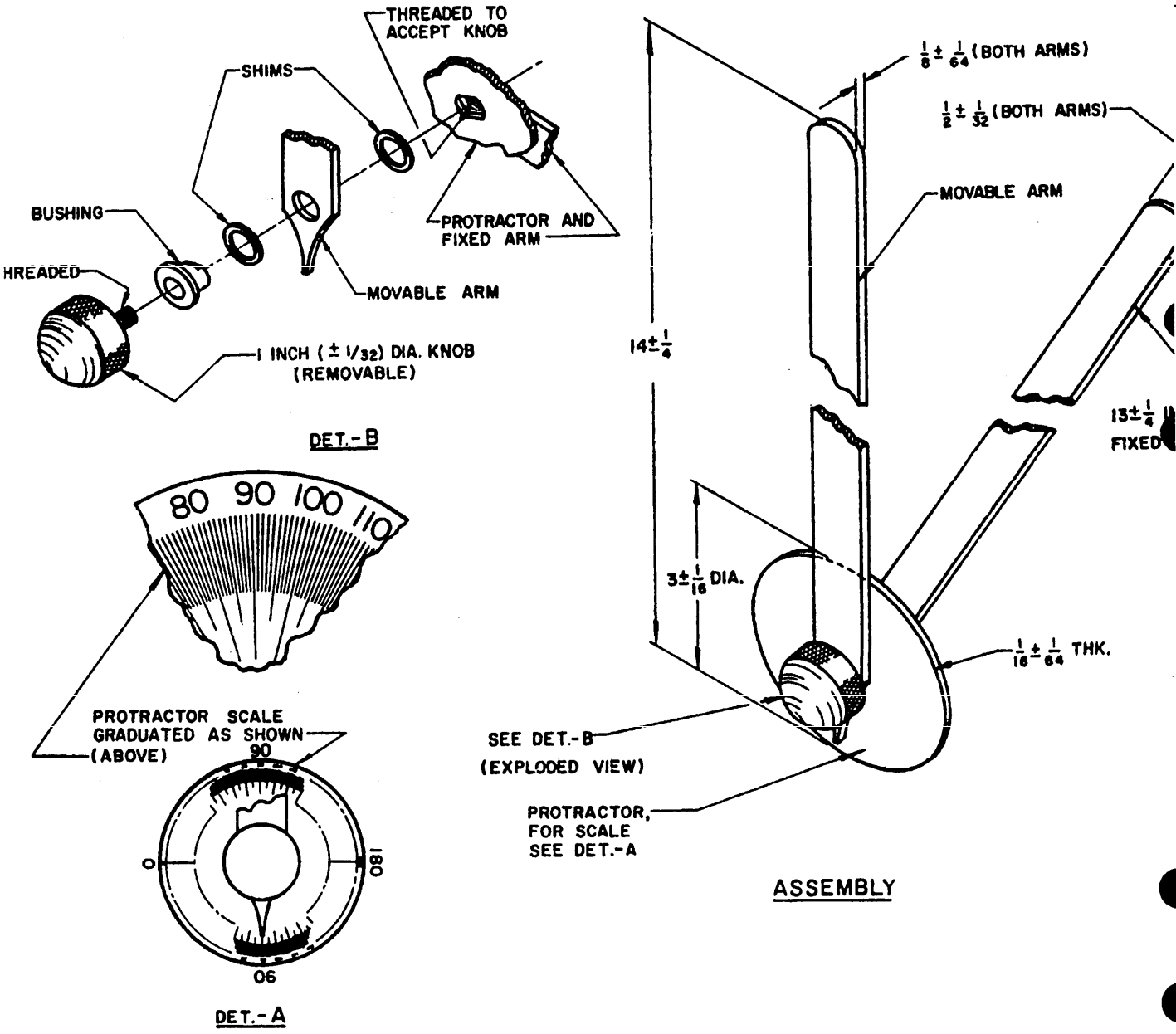


FIG.-1

3.2. Style, design, and dimensions. The goniometer shall be of the Conzett design, suitable for use in measuring the degree of flexion and extension of joints. The instrument shall consist of an arm permanently fixed to a protractor and a movable arm attached to the protractor by means of a removable knob, bushing, and two shims. When holding the assembled goniometer by one arm (with knob tightened by hand), the other arm shall not move on its own accord. The goniometer shall be in accordance with figure 1.

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3.3 Finish. The goniometer shall have a mirror, or mirror and satin, finish conforming to the requirements of Federal Specification GG–I–526.

3.4 Additional requirements. Except as specified herein, the goniometer shall comply with the applicable requirements and tests of Federal Specification GG–I–526. The requirements of that specification for micrometallographic examination shall not apply.

3.5 Workmanship. Workmanship shall be first class throughout. Goniometers shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* The inspection lot shall be determined by the cognizant government inspector and shall be based on the unit of product being one goniometer (end item).

4.2.2 *Sampling for examination.* A random sample of goniometers shall be selected from each lot offered to the Government in accordance with Military Standard MIL–STD–105. Examination for dimensions shall be in accordance with Military Standard MIL–STD–105, and as indicated in table 1.

4.2.3 *For tests.* Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105, and as indicated in table I.

4.2.4 *Raw materials.* Four ounces of each lot of each metal used in the construction of the goniometers shall be selected for chemical analysis.

TABLE I.—*Sampling criteria*

Component	Characteristic	Req.	Test procedure	Unit of product	Insp. level	AQL (percent defective)
Protractor, arms, knob, and bushing	Hardness	GG–I–526	GG–I–526	1—gon.	L3*	**
	Boil test	GG–I–526	GG–I–526	1—gon.	L3*	**
	Dimensions	3.2	.....	1—gon.	L3	4.0

* A minimum of three (3) units shall be tested regardless of lot size.
** Acceptance number shall be zero for all sample sizes.

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4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The goniometers shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

For civil agency procurement, Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.

5.1 Packaging and packing.

5.1.1 *Unit of issue.* One goniometer, as specified, constitutes one unit of issue.

5.1.2 *Packaging quantities.* The number of units of issue indicated in table II shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE II.—*Packaging quantities*

Unit package	Intermediate package	Exterior container
1 unit	Not required	48 units

5.1.2.1 *Packing variation permitted.* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.* Packaging and packing shall conform to the requirements of Military Specification MIL–P–12902, Procedure Code No. 35.

5.1.4 *Level B.* Packaging and packing shall conform to the requirements of Military Specification MIL–P–12902, Procedure Code No. 21.

5.1.5 *Level C.* The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking—nonmilitary.

5.2.1 Unless otherwise specified in the contract or order, unit packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3 Marking—military.

5.3.1 *Interior packages.* Marking shall conform to the requirements of Military Specification MIL–P–12902, except that the ASMPA stock number will not be required.

5.3.2 *Exterior container.* Exterior container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1).

(1) Level B preservation and packaging is intended to provide economical but limited protection, and should be specified only when it is determined the goniometer will be held in covered storage for not more than 12 months from date of initial packaging.

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6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Transportation description. Transportation description and minimum weight applicable to the commodity are:

Rail:

Surgical instruments, not otherwise indexed by name.
Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Surgical instruments, not otherwise indexed.
Truckload minimum weight 10,000 pounds, subject to Rule 34, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

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